



Dell Canvas tablet

Dell's new Canvas is a 27-inch tablet for creative professionals and a strong competitor for Microsoft's Surface Studio. <http://dgit.in/CanvasDell>



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Bazaar

Intel Core i7 7700K

The key word is optimisation

Anounced back in August 2016, the Kabylake microarchitecture is a follow up to the Skylake microarchitecture which was released about a year before that in August 2015. Departing from Intel's "Tick-Tock" cadence cycle, Kabylake forms a new step which Intel calls *Optimize* – a part of their new "process-architecture-optimization" cycle. So from the get go, you shouldn't expect tremendous gains from Kabylake, it is Intel's way of ironing out the design before the entire 14nm microarchitecture is shrunk down to 10 nm manufacturing process. And this "process" step will arise as Intel Cannonlake later this year.

There have been rumours that Intel might opt for a second optimisation step in the form of Coffee Lake before moving onto Cannonlake. This second level of "optimization" is supposedly due to issues with the 10 nm manufacturing process and to compete with AMD's Ryzen. Kabylake processors bring with it, a few key changes to the platform, including the new 200 series Union Point chipset among others. Yes, that means there shall be new motherboards.

Kabylake is how Intel hopes to materialise what it calls the "Immersive Internet". The improved IGP on Kabylake goes a long way towards making this a possibility. And that's by making 4K content consumption easier, helping bring popular Esports titles to each and every one along with VR and 360 degree video playback. While Kabylake is primarily an optimised version of Skylake, there are some noteworthy improvements to the entire platform overall:

1. The new 200-series Union Point chipset is compatible with Skylake CPUs and Kabylake CPUs are compatible with the older Sunrise Point 100-series chipsets.
2. Greater clock speeds are a given. Kabylake's flagship Core i7-7700K is clocked at 4.2 GHz with Turbo Boost propelling it to 4.5 GHz.
3. Kabylake will feature Intel HD Graphics 630 which has a peak clock speed of 1.15 GHz. HD Graphics 630 supports full hardware based VP9/



Price
₹28,999

HEVC decoding, boasting of handling 4K 60FPS 10-bit with ease.

4. Kabylake will support Optane, an upcoming memory technology set to hit the markets towards the end of 2017. Another name for the same is 3D Xpoint "3D-cross-point". It boasts 10x lower latency, 3-4x improvement in read and write speeds compared to NAND and a packing density that's four times greater than DRAM.
5. While the CPU will still provide for 16 PCI Express 3.0 lanes, the PCH will now provide 24 PCI Express 3.0 lanes which is an improvement over the 20 lanes that 100-series Sunrise Point chipsets used to provide.

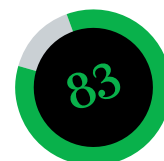
6. Interestingly, Kabylake will be the first micro-architecture since the Core i series debuted to have an unlocked Core i3 processor. The Core i3-7350K is a 2C/4T SKU with a base clock of 4.2 GHz, there's no boost on Core i3.

7. Overclocking on Kabylake has been made a lot more simpler, especially with the voltage levels. Even the instability that arose due to the AVX instruction set being called has been worked on to allow for a more stable higher OC frequency.

Performance of a different kind

With performance, the 7700K excels at a few tests but overall, the increment isn't that impressive. In CineBench the score difference was just 2 units in favour of the 7700K. However, when you move towards utilising the improved IGP you see a marked difference. Transcoding a 4K video was 9.8 per cent faster. On the 6700K with no discrete graphics card connected, we managed to get 45 FPS in GTA 5 and on the 7700K that went up to 54 FPS. We tried overclocking the CPU on air but were only able to complete benchmarks at 4.8 GHz. Overall, the CPU is the best performer in the high-end segment and we are really excited to see how the competition acts with the release of Ryzen right on the horizon. If you already have a 4790K or a 6700K, it's best to skip this one and wait for Cannonlake.

Mithun Mobandas



Performance.....90
Value for money.....75

Specifications

Clock Speed: 4.2 GHz;
Turbo frequency: 4.5 GHz; **Cores/Threads** - 4/8;
Lithography: 14nm; **TDP:** 91W; **IGP:** Intel HD 630;
IGP Frequency: 350 MHz;
IGP Max Freq.: 1.15 GHz.

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